Domácí úkol 07

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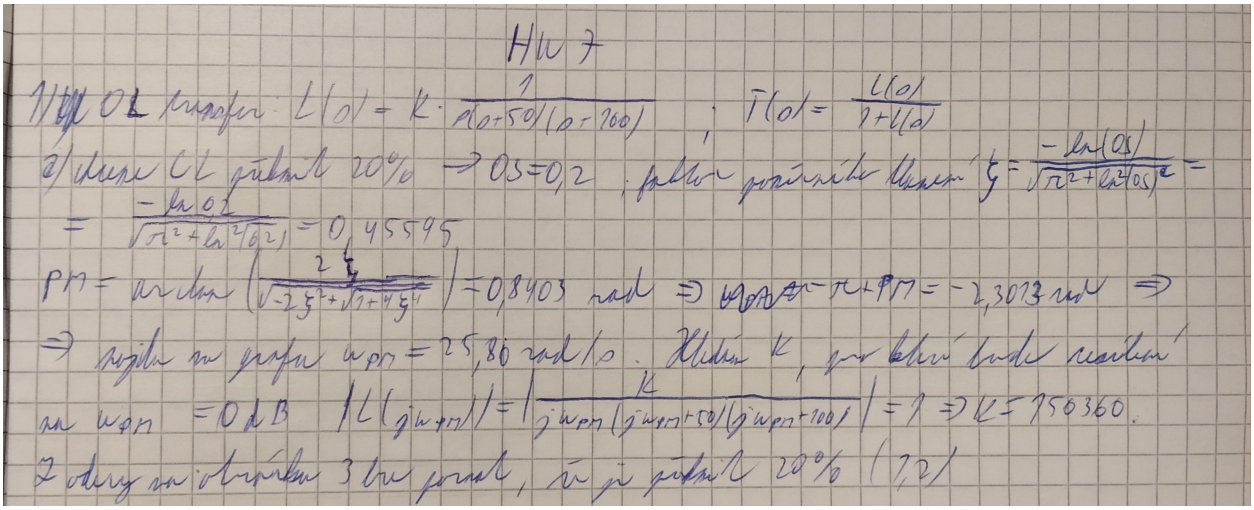
1. Příklad 1



HW 7

1) OL transfer: $L(s) = K \cdot \frac{1}{s(s+50)(s+100)}$; $T(s) = \frac{L(s)}{1+L(s)}$

2) chceme CL překmit 20% $\Rightarrow OS = 0{,}2$; faktor potřebného tlumení $\zeta = \frac{-\ln(OS)}{\sqrt{\pi^2 + \ln^2(OS)}} =$

$= \frac{-\ln 0{,}2}{\sqrt{\pi^2 + \ln^2(0{,}2)}} = 0{,}45595$

$PM = \arctan\left(\frac{2\zeta}{\sqrt{-2\zeta^2 + \sqrt{1+4\zeta^4}}}\right) = 0{,}8403$ rad $\Rightarrow$ ~~$\omega_{PM} = -\pi + PM$~~ $= -2{,}3013$ rad $\Rightarrow$

$\Rightarrow$ najdu na grafu $\omega_{PM} = 25{,}80$ rad/s. Hledám K, pro které bude nastavení na ω_{PM} = 0 dB $|L(j\omega_{PM})| = \left|\frac{K}{j\omega_{PM}(j\omega_{PM}+50)(j\omega_{PM}+100)}\right| = 1 \Rightarrow K = 750360$.

Z obrázku 3 lze poznat, že je překmit 20% (1,2)

Figure 1: zápisky 1

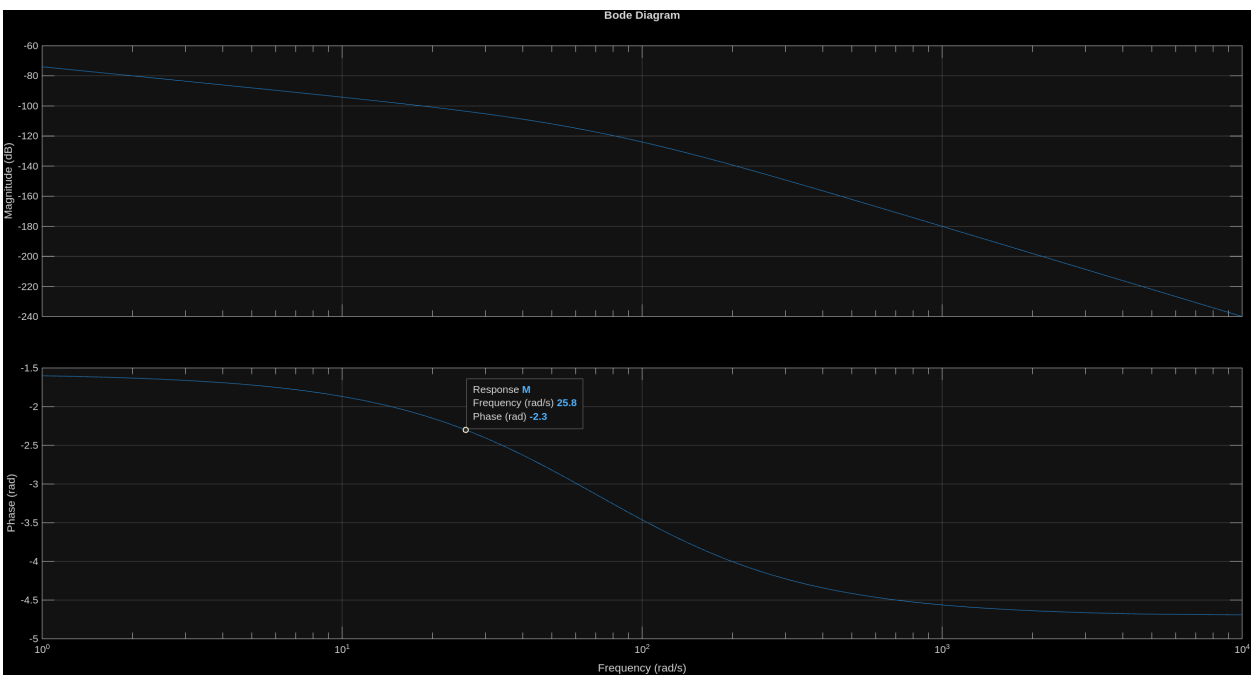


Figure 2: bode

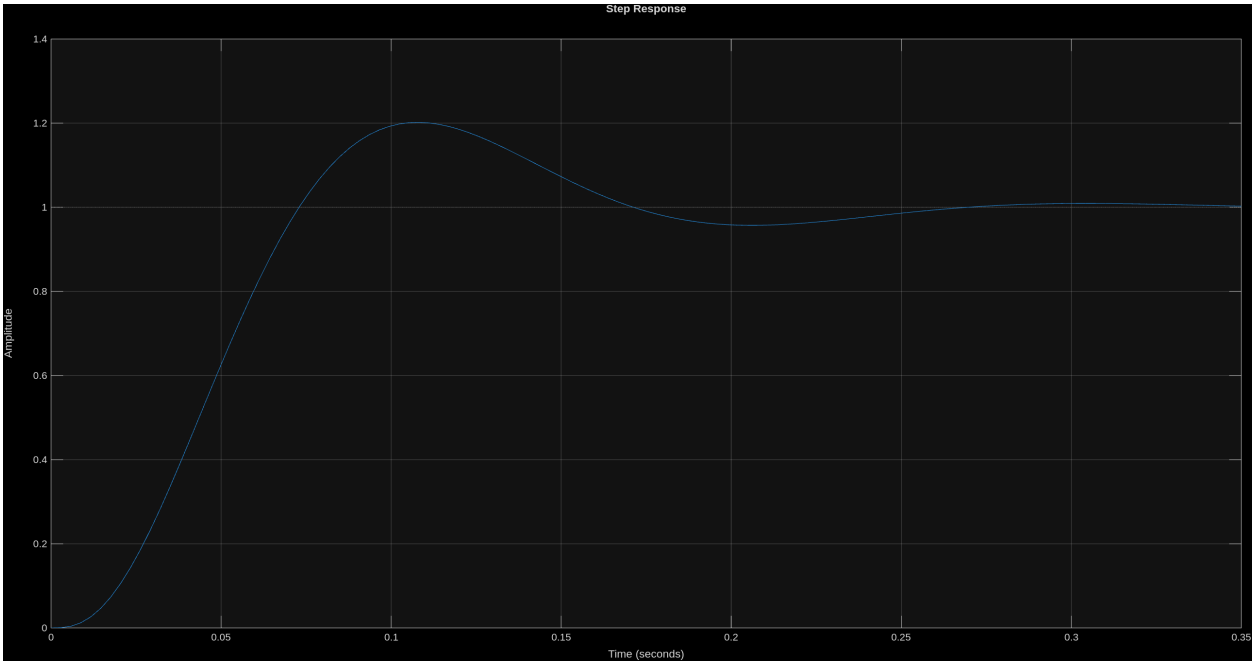
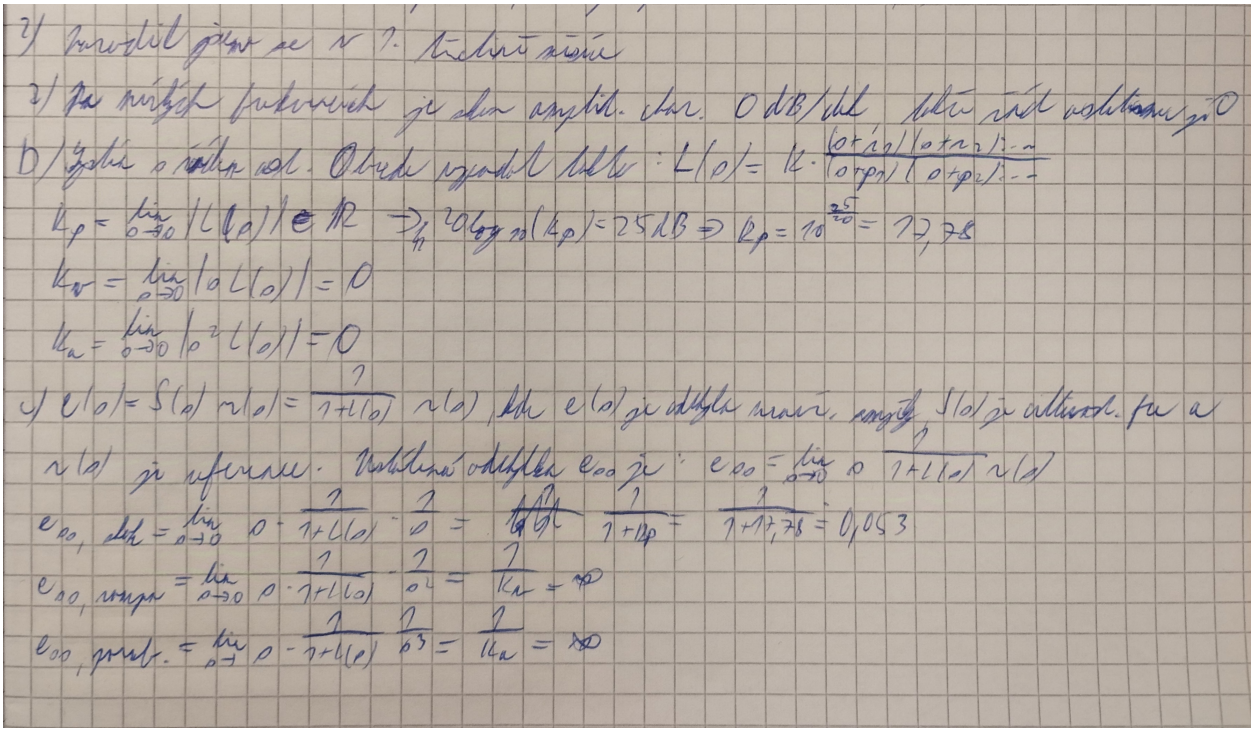


Figure 3: step response

2. Příklad 2



2) Zakreslit jako se v 1. kroku výše

2) Pro nízké frekvence je sklon amplit. char. 0 dB/dek, fáze začíná od nulového ~~0~~

b) Zjistit o kolik roste. Odtud vyjádřit koef.: $L(s) = k \cdot \frac{(s+z_1)(s+z_2)\dots}{(s+p_1)(s+p_2)\dots}$

$k_p = \lim_{s \to 0} |L(s)| \in \mathbb{R} \Rightarrow 20\log_{10}(k_p) = 25\,dB \Rightarrow k_p = 10^{\frac{25}{20}} = 17{,}78$

$k_v = \lim_{s \to 0} |sL(s)| = 0$

$k_a = \lim_{s \to 0} |s^2 L(s)| = 0$

c) $e(s) = S(s)\,r(s) = \frac{1}{1+L(s)}\,r(s)$, kde $e(s)$ je odchylka regulované veličiny, $S(s)$ je citlivost. fce a $r(s)$ je reference. Ustálená odchylka e_∞ je: $e_\infty = \lim_{s \to 0} s \frac{1}{1+L(s)}\,r(s)$

$e_{\infty,\,skok} = \lim_{s \to 0} s \cdot \frac{1}{1+L(s)} \cdot \frac{1}{s} = \frac{1}{1+k_p} = \frac{1}{1+17{,}78} = 0{,}053$

$e_{\infty,\,rampa} = \lim_{s \to 0} s \cdot \frac{1}{1+L(s)} \cdot \frac{1}{s^2} = \frac{1}{k_v} = \infty$

$e_{\infty,\,parab.} = \lim_{s \to} s \cdot \frac{1}{1+L(s)} \frac{1}{s^3} = \frac{1}{k_a} = \infty$

Figure 4: zápisky 2